FINAL REPORT

FISH ACUTE TOXICITY TEST

AS PER OECD 203

FOR TRANSOL SYNTH

SUBMITTED BY

SAVITA POLYMERS

ENVIRONMENTAL DIVISION LABORATORY, MUMBAI

INTERTEK INDIA PRIVATE LIMITED

Client: SAVITA POLYMERS

Sample registration date: 24/03/2023

Analysis starting date: 24/03/2023 (Pre conditioning) **Analysis completed on:** 10/05/2023

Name of product: TRANSOL SYNTH 100 (Laboratory Ref. no. 16425905)

Quantity received and packing: 200 gm in 250 ml PE can

Sample details: TRANSOL SYNTH 100

Test Required: Fish Acute Toxicity test as per OECD 203

Sampling done by: Sample not drawn by Intertek

Report No: MUM/001110/2023

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request

LABORATORY

Testing as presented in this report was conducted by Environmental division of Intertek India Private Limited. The testing facility is located at F wing, 2nd Floor, Chandivali Saki vihar Road, Andheri (East), Mumbai – 400 072, India.

SAMPLE RECEIPT

The sample was received on 24/03/2023 at the Intertek testing facility. The package was received by courier. Samples were at ambient temperature in good condition with no evidence of damage or contamination.

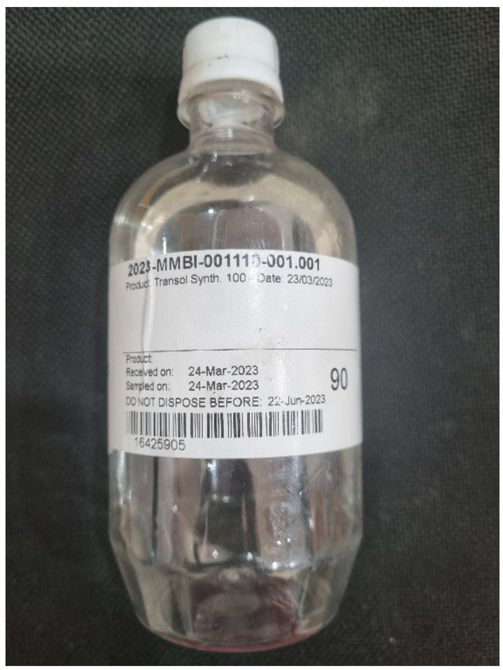

Figure 1: Test Sample

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request

PROJECT DESCRIPTION:

TRANSOL SYNTH 100 sample was submitted by **SAVITA POLYMERS** for studying the Fish Acute Toxicity Test as per OECD 203. The fish are exposed to the test chemical for a period of 96 hours, under either static, semi-static or flow-through conditions. Mortalities and visible abnormalities related to appearance and behaviour are recorded.

VALIDITY OF THE TEST

For a test to be valid, the following conditions should be fulfilled:

- in the control(s) (dilution water control, solvent control), the mortality should not exceed 10% (or one fish, if fewer than 10 control fish are tested) at the end of the exposure;
- the dissolved oxygen concentration should be ≥60% of the air saturation value in all test vessels throughout the exposure;
- analytical measurement of test concentrations is compulsory.

EXPERIMENTAL DESIGN:

A) Test Vessels:

Any glass, stainless steel or other chemically inert vessels can be used.

B) Danio rerio (Zebrafish) :

Total length of test fish were 2-4 cm. They can be bred and cultivated in the laboratory, under disease- and parasite-controlled conditions, so that the test fish will be healthy and of known parentage. Fish were of same age and had normal appearance.

C) Reconstituted water (for test solution) :

Per liter of reconstituted water, it contains: 96.0 mg $NaHCO_3$, 123.0 mg $MgSO_4\ 7H_2O$, 4.0 mg KCl, 60.0 mg $CaSO_4\ 2H_2O$. The reconstituted water has been strongly exposed to air until 48 hours before the test day.

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request

D) Holding of fish :

All fish must be obtained and held in the laboratory for at least 9 days before they are used for testing. The first 48 hours constitute a settling-in period. Then, fish should be acclimatized for at least 7 days (48 hours settling-in + 7 days acclimatization = 9 days) in water similar to test water immediately before the start of the test. They must be held in water of the quality to be used in the test for at least seven days immediately before testing and under the following conditions:

- Light: 12-16 hours photoperiod daily.
- Temperature: 24 ± 2℃.
- Oxygen concentration: at least 80 per cent of air saturation value.
- Feeding: daily until 24 hours before the test is started.

Following a 48-hour settling-in period, mortalities are recorded, and the following criteria are applied: Mortalities of greater than 10 per cent of population in seven days: rejection of entire batch; Mortalities of between 5 and 10 per cent of population: acclimatization continued for seven additional days; Mortalities of less than 5 per cent of population: acceptance of batch.

E) Water :

Good quality natural water or reconstituted water is preferred. Waters with total hardness of between 40 and 250 mg $CaCO_3$ per liter, and with a pH 6.0 to 8.5 are preferable. The reagents used for the preparation of reconstituted water should be of analytical grade and the deionized or distilled water should be of conductivity equal to or less than 10 μScm-1. Dilution water is aerated prior to use so that dissolved oxygen concentration has reached saturation.

F) Test solutions :

Test solutions of chosen concentrations are prepared by dilution of a stock solution. The test should be carried out without adjustment of ph. If there is evidence of marked change in the pH of the tank water after addition of the test substance, it is advisable that the test be repeated, adjusting the pH of the stock solution to that of the tank water before addition of the test substance. This pH adjustment should be made in such a way that the stock solution concentration is not changed to any significant

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request

extent and that no chemical reaction or precipitation of the test substance is caused. HCl and NaOH are preferred.

G) Procedure :

1. Conditions of exposure.

- Duration: 96 hours
- Loading: maximum loading of 1.0 g fish/liter
- Light: 16 hours photoperiod daily
- Temperature: 22 ± 2℃
- Oxygen concentration: not less than 60 per cent of the air saturation value
- Feeding: none
- Disturbance: disturbances that may change the behavior of the fish should be avoided

2. Number of fish: 7 fish were used at each test concentration and in the controls.

3. Test concentrations: For definitive test with fish,5 concentrations were performed.

4. Controls Solvent control or dilution water control can be used as blank.

OBSERVATIONS

The fish were inspected at least after 24, 48, 72 and 96 hours. Fish were considered dead if there was no visible movement (e.g., gill movements) and if touching of the caudal peduncle produces no reaction. Dead fish are removed when observed and mortalities are recorded. Observations at three and six hours after the start of the test are desirable. Records were made on visible abnormalities (e.g., loss of equilibrium, swimming behavior, respiratory function, pigmentation, etc.). Measurement of pH, dissolved oxygen and temperature should be carried out at least daily.

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request

RESULTS

Water quality characteristics (pH, oxygen concentration, and temperature)

Table 1. pH of test solutions

Hours / Concentration	0	24	48	72	96
Control	7.50	7.50	7.42	7.40	7.40
25 mg/l	7.36	7.30	7.28	7.24	7.20
50 mg/l	7.16	7.12	7.10	7.06	7.04
75 mg/l	7.08	7.00	6.92	6.90	6.88
100 mg/l	7.00	6.90	6.90	6.88	6.84

Table 2. Oxygen concentration of test solutions

Hours / Concentration	0	24	48	72	96
Control	6.70	6.60	6.40	6.30	6.20
25 mg/l	6.60	6.50	6.10	5.80	5.00
50 mg/l	6.20	6.00	5.70	5.36	5.10
75 mg/l	6.15	5.90	5.60	5.30	4.48
100 mg/l	6.10	5.60	5.30	4.90	4.00

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request

Table 3. Temperature of test solutions

Hours / Concentration	0	24	48	72	96
Control	22.7	22.6	22.4	22.5	22.2
25 mg/l	22.3	22.4	22.7	22.5	22.1
50 mg/l	22.4	22.6	22.9	22.7	22.4
75 mg/l	22.3	22.4	22.8	23.0	22.6
100 mg/l	22.2	22.3	22.7	22.9	22.6

Table 4. Definitive test

Hours / Concentration	0	24	48	72	96	Total Mortality
Control	0	0	0	0	0	0
25 mg/l	0	0	0	0	0	0
50 mg/l	0	0	0	0	0	0
75 mg/l	0	0	0	0	0	0
100 mg/l	0	0	0	0	0	0

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request

CONCLUSION

According to Table 4, all the 5 concentrations caused no mortality within the period of the test.

-- End of Report ------------------------------

Authorized Signatory

Jayashree

Intertek India Pvt. Ltd
AA division, Chandivali

Jayashree Acharya
Assistant Manager-Environment laboratory

Note : The results reported relate to the sample tested only. The Test Certificate shall not be reproduced in full or part without the written permission of INTERTEK. The sample(s) has been not drawn/sampled by INTERTEK LAB. The reported result(s) provide no warranty or verification on the sample(s) representing any specific goods and / or shipment and only relate to the sample(s) as received and tested. This report was prepared solely for the use of the client named in this report. **Intertek** disclaims any and all liability for damage or injury which results in the use of the information contained herein and accepts no responsibility for any loss, damage or liability suffered by a third party as a result of any reliance upon or use of this report. All jobs are performed as per Intertek terms & Conditions available at **htpp://www.intertek.com/terms**/or can be made available on request